Stability-Penalized Split Selection for Decision Tree Classification: A Reproducible Multi-Dataset Evaluation

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Abstract

The CART decision tree is fast and interpretable, but its structure is unstable: small changes in the training data can alter the selected split and, ultimately, the predictions. This study evaluates a very simple candidate formulation, named *Stability-Penalized Decision Tree* (SPDT), in which each candidate split is scored as Gini gain minus a weighted penalty on the variance of that gain over bootstrap resamples of the node data: `Score(s) = Gain(s,D) − lambda*Var_b[Gain(s,D^(b))]`. A literature search up to 22 October 2026 found no formula-level equivalent of this criterion, but the tree-stability research space is crowded; the formulation is therefore positioned as a candidate, not as a novelty claim. The evaluation uses 8 UCI tabular classification datasets with repeated nested cross-validation (5 folds x 3 repeats on three seeds), hyperparameter tuning in the inner loop, and Macro-F1 as the primary metric. Results: SPDT versus custom CART (lambda=0) records 6 wins, 0 ties, and 2 losses with a mean Macro-F1 difference of +0.0058. The Friedman test gives chi^2(2)=4.0000, p=0.1353, Kendall's W=0.2500 – insufficient evidence to reject the null hypothesis, so confirmatory post-hoc comparisons were not run. Ablations show that the penalty strength lambda and the number of resamples B barely affect Macro-F1, tree-depth constraints matter far more, and the prediction-stability measure of SPDT is slightly worse than that of custom CART at roughly 40x the fitting time. These findings are an honest null result: across the eight small-to-medium datasets tested, the bootstrap gain-variance penalty provides no benefit sufficient to outweigh the simplicity of CART, without ruling out benefits in other data regimes.

I. Introduction

Decision trees are nonparametric classifiers that recursively partition the feature space. The CART split criterion selects the candidate (feature, threshold) that maximizes the reduction of Gini impurity (Breiman et al. 1984; Quinlan 1986). However, tree structure is known to be unstable: small perturbations of the training data can change the selected split (Li and Clarke 2002), and consequently the predictions – the very phenomenon behind *bagging* and *random forests* (Breiman 1996, 2001; Dietterich 2000).

Various approaches have reduced this instability at different levels: significance filtering, structural restriction, ensemble aggregation, or stable model selection (Briand et al. 2009; Lee et al. 2024; Mirzamomen and Kangavari 2017; Wang et al. 2017). What is rarely tested directly is the narrower question:

> **Can a bootstrap variance penalty on the split gain, subtracted directly from the split score, improve out-of-sample Macro-F1 and/or prediction stability compared with standard CART under repeated resampling, at acceptable runtime cost?**

Null hypothesis: no systematic improvement over the baseline across independent datasets. No hypothesis is made that SPDT *must* win.

The claimed contributions are limited to: (i) an explicit candidate formulation; (ii) a reproducible implementation and audit; (iii) a leakage-free nested-CV evaluation; (iv) honest ablation and statistical analysis of both positive and negative outcomes.

II. Related Work

Decision tree stability. Structural tree instability was formalized by Li and Clarke (Li and Clarke 2002); quantitative stability and tree-similarity measures were developed, among others, by Turney (Turney 1995) and Briand et al. (Briand et al. 2009), who also evaluated bootstrap-based node-level stabilization (the NLS/Dannegger lineage). Lee et al. (Lee et al. 2024) proposed ANLS, which runs bootstrap iterations at each node and selects the most frequently chosen predictor – the closest resampling-per-node mechanism to this study, but its aggregation is frequency voting rather than a gain-variance penalty. Mirzamomen and Kangavari (Mirzamomen and Kangavari 2017) stabilize trees through multi-attribute split tests.

Alternative split criteria. Two-term Tsallis entropy criteria (Wang et al. 2017), label-noise-robust criteria (Yang et al. 2019), bias-corrected Gini split selection (Strobl et al. 2007), and Hoeffding-bound data-stream trees (Domingos and Hulten 2000) show that the "modify the split score" research space is active. Delise (Delise 2026) scores split

consistency across environments in boosting. Penalizing a resampled score by its variance is also known outside trees (the mean-variance pattern in model selection). None of the works we were able to verify contains the formula `Gain - lambda*Var_bootstrap[Gain]`; nevertheless, a negative search result is not proof of novelty, and the formulation remains a candidate.

III. Method

A. Proposed split criterion: For a node with dataset D (training portion only), with `Gini(D) = 1 - Sum_c p(c|D)^2` and split gain

```
Gain(s,D) = Gini(D) -
(|D_L|/|D|)*Gini(D_L) -
(|D_R|/|D|)*Gini(D_R),
```

B bootstrap resamples `D^(b)` are drawn from the node rows. For every valid candidate s:

`V(s) = Var_b[Gain(s,D^(b))]` (sample variance, ddof=1)

Main score: `Score(s) = Gain(s,D) - lambda*V(s)`.

Deterministic policies: candidate thresholds are midpoints between sorted distinct adjacent values; `min_samples_leaf`/`min_samples_split` are applied to the original node counts; an empty child partition inside a resample contributes 0; leaves predict the majority class with smallest-label tie-breaking; bootstrap seeds derive from `(random_state, node id)`. For lambda=0 the entire bootstrap computation is skipped and the estimator degenerates to the custom CART reference (verified identical to an independent reference implementation in unit tests). A separate exploratory variant (normalization by `V/(Gain^2+eps)`) is documented but is not the main method. An important limitation: V(s) is scale-dependent on the gain.

B. Evaluation design: Datasets: 8 UCI tabular classification datasets (Table I). Protocol: repeated stratified nested CV – outer 5 folds x 3 repeats for each of the seeds [42, 123, 2026] (45 evaluations per dataset-method, identical splits for all methods); inner stratified 3-fold tuning with deterministically derived seeds. Preprocessing is median imputation fitted on the training portion only (pipeline). Primary metric Macro-F1; secondary: balanced accuracy, accuracy, fit/predict time, node count, tree depth. Prediction stability is measured separately as mean pairwise disagreement among models trained on 10 subsamples (90%) of each outer training fold, evaluated on the same outer test fold – reported separately from accuracy.

C. Statistical analysis: The unit of inference is the dataset (averaged over 45 outer folds), not the fold. The three primary methods (sklearn CART, custom CART lambda=0, SPDT) are compared with the Friedman test using datasets as blocks; only when the omnibus test provides grounds for it are paired Wilcoxon signed-rank tests with Holm correction and rank-biserial effect sizes run. p >= 0.05 is never interpreted as evidence of equivalence. Random forest is included as a non-equivalent contextual benchmark and is not part of the primary omnibus test.

IV. Results and Discussion

Post-hoc not run: the Friedman omnibus test gave no grounds (p >= 0.05), per protocol.

Table: Pairwise (post-hoc) comparisons.

Paired summary SPDT vs custom CART: **6 wins / 0 ties / 2 losses**, mean Macro-F1 difference +0.0058.

Discussion. The results descriptively favor SPDT but do not reach confirmatory evidence. SPDT achieves the best mean rank (1.500 vs 2.000 for sklearn CART and 2.500 for custom CART) and wins on 6 of 8 datasets paired against custom CART, yet the Friedman test finds insufficient evidence of a difference (p = 0.1353), and the exploratory SPDT-custom CART comparison gives a raw p of 0.078 – it does not survive the 5% level and must not be read as significant. With only 8 datasets, statistical power is limited; per protocol, the results are reported as they stand and p >= 0.05 is not interpreted as equivalence.

A second important finding: the stability penalty does not reduce prediction disagreement – SPDT is in fact slightly higher (0.101) than custom CART (0.093) and sklearn CART (0.098). Random forest remains the most stable (0.027) through ensemble aggregation. The motivating promise of the method (better prediction stability) therefore does not materialize on this benchmark; a gain-variance penalty is no substitute for ensembling.

A third finding from the identical-split ablations: neither the penalty strength lambda nor the number of resamples B materially changes Macro-F1 (all lambda and B variants fall in the 0.859-0.861 range), whereas depth/leaf constraints matter far more (depth 3 drops to about 0.836). Inner tuning tends to select large lambda (10 in 64.7% of folds) and a small B (5 in 53.1%), indicating the penalty is often selected but has a small effect on accuracy. The cost is real: SPDT fitting time is roughly 40x that of custom CART. Overall, the effectiveness of the bootstrap gain-variance penalty is dataset-dependent and small in magnitude; simple impurity-based splitting remains preferable for these small-to-medium datasets.

V. Conclusion

This study evaluated the SPDT candidate formulation reproducibly, with leakage-free repeated nested CV, controlled ablation on identical splits, and dataset-level statistical analysis on 8 UCI datasets. Descriptively, SPDT achieved the best mean rank and won 6 of 8 paired comparisons against custom CART, but the Friedman test provided insufficient evidence

TABLE I: Dataset inventory.

Dataset	N	Features	Classes	Source
banknote	1372	4	2	ucimlrepo
glass	214	9	6	ucimlrepo
ionosphere	351	34	2	ucimlrepo
iris	150	4	3	ucimlrepo
seeds	210	7	3	uci_static_zip
sonar	208	60	2	ucimlrepo
wdbc	569	30	2	ucimlrepo
wine	178	13	3	ucimlrepo

TABLE II: Macro-F1 mean +/- sd (descriptive, fold-level) per dataset and method.

Dataset	cart_sklearn	cart_custom	spdt	random_forest	Delta (spdt-cart_custom)
banknote	0.982 +/- 0.008	0.981 +/- 0.007	0.981 +/- 0.007	0.994 +/- 0.004	+0.0004
glass	0.620 +/- 0.097	0.609 +/- 0.093	0.626 +/- 0.093	0.742 +/- 0.075	+0.0166
ionosphere	0.873 +/- 0.034	0.881 +/- 0.038	0.879 +/- 0.038	0.929 +/- 0.037	-0.0027
iris	0.949 +/- 0.033	0.945 +/- 0.038	0.944 +/- 0.037	0.955 +/- 0.034	-0.0008
seeds	0.897 +/- 0.032	0.906 +/- 0.036	0.913 +/- 0.034	0.923 +/- 0.038	+0.0066
sonar	0.719 +/- 0.065	0.706 +/- 0.073	0.725 +/- 0.061	0.833 +/- 0.065	+0.0189
wdbc	0.925 +/- 0.031	0.925 +/- 0.030	0.928 +/- 0.030	0.956 +/- 0.023	+0.0037
wine	0.905 +/- 0.047	0.905 +/- 0.044	0.909 +/- 0.046	0.981 +/- 0.024	+0.0040

TABLE III: Mean Macro-F1 rank.

Method	Mean rank
spdt	1.5000
cart_sklearn	2.0000
cart_custom	2.5000

TABLE IV: Dataset-level statistical tests.

Test	Statistic	df	p	#datasets	Kendall W	Decision
friedman	4.0000	2	0.1353	8	0.2500	no omnibus evidence; post-hoc not run

of a difference (chiˆ2(2)=4.0000, p=0.1353), so there is no confirmatory basis for claiming improvement. Moreover, the prediction-stability measure did not improve and was slightly worse, while compute cost rose roughly 40x. Under the tested conditions, the simplicity of CART remains preferable. Limitations: eight small datasets, Gini impurity only, a single disagreement-based stability proxy, a formula-level novelty claim that is not established, and the absence of the closest literature criteria (ANLS, minimax split) as comparators. The most valuable future work: direct comparison with node-level bootstrap stabilization from the literature, a theoretically grounded penalty normalization by gain scale, and evaluation on larger benchmarks.

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TABLE V: Computational cost.

Method	Fit mean (s)	Fit median (s)	Predict mean (s)	Mean nodes	Mean depth
cart_custom	0.5340	0.4163	0.0009	26.6722	5.1000
cart_sklearn	0.2693	0.2605	0.0010	27.4222	5.2917
random_forest	1.1169	1.0863	0.0573	nan	nan
spdt	21.4302	14.6491	0.0010	28.5944	5.4611

TABLE VI: Prediction stability (disagreement; not an accuracy metric).

Method	Mean disagreement	SD
cart_custom	0.0928	0.0853
cart_sklearn	0.0983	0.0885
random_forest	0.0271	0.0337
spdt	0.1014	0.0934

TABLE VII: Ablation on identical splits.

Variant	Mean macro F1	SD	Fit mean (s)	Mean nodes
A0_lam0	0.8599	0.1270	0.0201	37.6611
A1_lam0.1_B10	0.8598	0.1256	0.0860	37.6111
A1_lam1_B10	0.8601	0.1255	0.0862	37.5278
A2_B20	0.8601	0.1251	0.1136	37.4556
A2_B5	0.8613	0.1243	0.0711	37.4667
A3_d3_leaf1	0.8357	0.1592	0.0371	12.8722
A3_d3_leaf5	0.8353	0.1546	0.0373	11.8611
A3_d8_leaf1	0.8592	0.1272	0.0821	35.2500
A3_d8_leaf5	0.8521	0.1381	0.0820	24.7389

TABLE VIII: SPDT hyperparameter selection frequency (45 outer folds per dataset).

parameter	value	frequency	share
max_depth	5.0000	140	0.3889
max_depth	8.0000	102	0.2833
max_depth	3.0000	92	0.2556
min_samples_leaf	1.0000	201	0.5583
min_samples_leaf	2.0000	97	0.2694
min_samples_leaf	5.0000	62	0.1722
min_samples_split	2.0000	252	0.7000
min_samples_split	5.0000	69	0.1917
min_samples_split	10.0000	39	0.1083
n_bootstrap	5.0000	191	0.5306
n_bootstrap	20.0000	87	0.2417
n_bootstrap	10.0000	82	0.2278
penalty_lambda	10.0000	233	0.6472
penalty_lambda	0.0000	61	0.1694
penalty_lambda	0.0100	41	0.1139
penalty_lambda	1.0000	22	0.0611
penalty_lambda	0.1000	3	0.0083

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